AGRICULTURAL ENGINEERING

Agricultural engineering courses focus on applying engineering principles to food and fiber production, covering soil/water conservation, farm machinery, irrigation, and post-harvest technology. The curriculum typically spans 4-5 years, combining engineering mechanics with agricultural science and practical industrial attachments to prepare students for mechanization and management roles.

Core Curriculum Components

- **Farm Power and Machinery:** Principles of tractors, internal combustion engines, and agricultural machinery design.
- **Soil and Water Engineering:** Irrigation design, drainage systems, soil mechanics, and hydrology
- **Agricultural Processing & Food Engineering:** Post-harvest technologies, crop drying, storage, and processing.
- **Agricultural Structures:** Design of greenhouses, farm buildings, and rural infrastructure.
- **Engineering sciences:** Thermodynamics, Fluid Mechanics, Engineering drawing, and Engineering Mathematics.
- **Environmental & Bio-systems:** Waste management, renewable energy, and environmental impact assessment.

Typical program Structure

Duration: Certificate course range from short courses to one year

Practical Training: Includes a compulsory external industrial attachment, typically lasting eight weeks.

Specialization: Studies in precision agriculture, machine design, or post-harvest technology
Important: For further details contact the college management

Short Courses

1. **Soap Making**
    - **Safety and Basics**: understanding the science of saponification, handling lye safely, and essential equipment

- ## Soap making techniques:
    - **Cold press:** Mixing oils and lye for a slow- cured, custom soap

  - **Hot press:** Cooking the soap for faster saponification and rustic look
  - **Melt – and - Pour:** Designing sops using pre-made bases
  - **Liquid soap making:** Creating liquid and cream soaps
- **Ingredients and Formulation:** Choosing oils, fats and butters, and learning to calculate recipes
- **Customization:** Incorporating natural colorants, herbs, botanical additives, and essential oils for scent.
- **Design and Troubleshooting:** Learning swirl techniques and fixing common issues like seized soap or false trace.
- **Business Basics:** Labelling, packaging, and local regulations for selling.

Agricultural Value Chain Training Skills

- **Introduction to Value Chains:** Definition of value chain actors and stages (input supply, production, processing, and distribution).
- **Value Chain Mapping and Analysis:** Methodologies for mapping chains, identifying nodes, assessing competitiveness, and diagnosing constraints.
- **Market Linkages and Analysis:** Strategies for connecting farmers to markets, understanding demand, and establishing contract farming.
- **Value Addition and Upgrading:** Techniques for product quality improvement, processing, and developing certification standards.
- **Value Chain Governance and Finance:** Understanding power dynamics within Chains, managing relationships, and accessing financial services.
- **Sustainability and Gender:** Integrating social inclusion, gender mainstreaming, and environmental sustainability in value chains.

Key Focus Areas

- **Development of Inclusive Business Models:** Ensuring smallholder farmers benefit from market participation
- **Risk Management:** Identifying and overcoming challenges such as logistics gaps, climate risks, and market fluctuations.
- **Practical Tools:** Utilizing analytical techniques for mapping and assessing chain performance.

N/B: These courses are intended for professionals in agricultural development, agribusiness managers, and policy makers looking to improve market competitiveness and sustainability.